Day 1 – Networking Fundamentals (Interview Notes)

Study Time: 2.5 Hours

Activity	Duration
Theory	60 min
Linux Hands-on	30 min
Practical	30 min
Revision	15 min
Interview Questions	15 min

Learning Objectives

After completing Day 1, you should be able to explain:

- What is a computer network?
- Why networking is required
- Types of networks
- Client-Server architecture
- Peer-to-Peer architecture
- What is a NIC?
- MAC Address
- IP Address
- Hostname
- Port Number
- Protocol
- Packet
- Frame
- Segment
- Datagram

You should also know how to find your IP address and MAC address in Linux.

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1. What is Computer Networking?

Definition

A **computer network** is a collection of two or more devices connected together so they can exchange data and share resources.

Resources include:

- Files

- Printers
- Internet
- Applications
- Databases
- Cameras
- IoT Devices

Example

```
Laptop  --------\
                 \
Phone ---------- Router -------- Internet
                 /
Desktop --------/
```

Every device connected to the network is called a **Host**.

Why Do We Need Networking?

Without networking:

- No Internet
- No Email
- No WhatsApp
- No Video Calls
- No Cloud Storage
- No Online Banking

Networking enables:

- Communication
- Resource Sharing
- Remote Access
- Centralized Management
- Data Transfer

Real-Life Example

Imagine a company.

100 employees need:

- Printer
- Internet
- Database
- Email

Instead of 100 printers,

All employees connect to one network printer.

Networking reduces cost and improves efficiency.

2. Types of Networks

LAN (Local Area Network)

Definition

A LAN connects computers within a limited geographical area.

Examples

- Home
- School
- Office

```
PC1 -----\
          \
PC2 ---- Switch
          /
PC3 -----/
```

Characteristics

- High Speed
- Low Latency
- Private Network

Interview Question

Q: Give an example of LAN.

Answer:

Office network.

MAN (Metropolitan Area Network)

Definition

A MAN connects multiple LANs inside a city.

Example

```
Office A ------ ISP ------- Office B
```

Coverage

- City
- Campus

WAN (Wide Area Network)

Definition

A WAN connects computers across countries.

Example

Internet

```
India ------------- USA
```

Characteristics

- High latency
- Huge coverage

Comparison

Feature	LAN	MAN	WAN
Coverage	Building	City	Country/World
Speed	High	Medium	Lower than LAN
Cost	Low	Medium	High
Example	Office	University Campus	Internet

---

3. Network Components

Every network contains:

```
Computer

↓

NIC

↓

Switch

↓

Router

↓

Internet
```

---

4. What is NIC?

NIC

= Network Interface Card

It is hardware that connects a computer to the network.

Without NIC

No network communication is possible.

Example

```
+------------------+

Motherboard

[ NIC ]

|

RJ45 Cable
```

Modern laptops have

- Ethernet NIC
- Wi-Fi NIC

Interview Question

What is the purpose of NIC?

Answer

It provides an interface between the operating system and the physical network.

---

5. MAC Address

Definition

MAC

Media Access Control Address

A unique hardware address assigned to every NIC.

Example

```
08:00:27:13:69:77
```

Length

48 bits

6 bytes

Example Format

```
AA:BB:CC:DD:EE:FF
```

Properties

- Unique
- Burned into hardware (though often configurable in software)
- Used inside LAN

How to Find MAC Address

```
ip link
```

Example

```
2: enp0s3:

link/ether 08:00:27:13:69:77
```

MAC Address

```
08:00:27:13:69:77
```

Interview Question

Why is MAC required?

Answer

The switch forwards Ethernet frames using MAC addresses.

---

6. IP Address

Definition

IP Address uniquely identifies a device in a network.

Example

```
192.168.1.10
```

Unlike MAC,

IP can change.

Example

Home Wi-Fi

```
192.168.1.10

↓

Office Wi-Fi

↓

10.1.10.20
```

IPv4

32 bits

Example

```
192.168.1.100
```

IPv6

128 bits

Example

```
2001:db8::1234
```

Interview Question

Difference between MAC and IP?

MAC	IP
Hardware Address	Logical Address
Permanent (typically)	Can Change

MAC	IP
Layer 2	Layer 3

---

7. Hostname

Hostname

Human-readable computer name.

Example

```
deepak-pc
```

Find hostname

```
hostname
```

---

8. Port Number

A port identifies a specific application or service running on a host.

Think of it like this:

- IP address = Apartment building
- Port = Apartment number

```
IP = 192.168.1.10

Port = 80
```

Different applications use different ports:

- Web server → Port 80 (HTTP)
- Secure web server → Port 443 (HTTPS)
- SSH → Port 22

Types of Ports

Range	Description
0–1023	Well-known ports

Range	Description
1024–49151	Registered ports
49152–65535	Dynamic/Ephemeral ports

Interview Question

Can two applications use the same port?

Answer

Not simultaneously on the same IP and protocol unless specific techniques (such as `SO_REUSEPORT`) are used. Normally, only one process can bind to a given IP/port/protocol combination.

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9. Protocol

A protocol is a set of rules that define how devices communicate.

Examples:

- HTTP – Web browsing
- HTTPS – Secure web browsing
- FTP – File transfer
- SMTP – Sending email
- DNS – Name resolution
- SSH – Secure remote login

Without protocols, devices would not understand each other's messages.

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10. Data Units

As data moves through the network stack, it is encapsulated into different units.

Layer	Data Unit
Application	Data
Transport (TCP)	Segment
Transport (UDP)	Datagram
Network	Packet
Data Link	Frame

Layer	Data Unit
Physical	Bits

Visualization:

```
Application Data
        |
        ▼
TCP Segment
        |
        ▼
IP Packet
        |
        ▼
Ethernet Frame
        |
        ▼
Bits on the wire
```

11. Basic Linux Networking Commands

`hostname`

Purpose:

Display the system's hostname.

Syntax:

```
hostname
```

Example:

```
deepak-pc
```

`hostname -I`

Purpose:

Display assigned IP address(es).

Syntax:

```
hostname -I
```

Example:

```
192.168.1.10
```

`ip addr`

Purpose:

Show all network interfaces and IP addresses.

Syntax:

```
ip addr
```

Look for:

- Interface name (e.g., `eth0` , `enp3s0` , `wlan0`)
- `inet` (IPv4 address)
- `inet6` (IPv6 address)

`ip link`

Purpose:

Display network interfaces and MAC addresses.

Syntax:

```
ip link
```

`ping`

Purpose:

Test network connectivity using ICMP Echo Request/Reply.

Syntax:

```
ping 8.8.8.8
```

Useful options:

- `-c 4` → Send only 4 packets.
- `-i 1` → Interval of 1 second between packets.

Example:

```
ping -c 4 google.com
```

Hands-on Exercise

1. Display your hostname:

```
hostname
```

2. Display your IP address:

```
hostname -I
```

3. Show all interfaces:

```
ip addr
```

4. Find your MAC address:

```
ip link
```

5. Test loopback connectivity:

```
ping -c 4 127.0.0.1
```

6. Test internet connectivity:

```
ping -c 4 8.8.8.8
```

7. Test DNS resolution:

```
ping -c 4 google.com
```

Observe the difference between pinging an IP address and a hostname.

Common Interview Questions

1. What is a computer network?

A group of interconnected devices that communicate and share resources.

2. What is the difference between LAN and WAN?

LAN covers a small area (e.g., an office), while WAN connects networks across large geographic areas (e.g., the Internet).

3. What is a NIC?

A hardware interface that allows a computer to connect to a network.

4. What is a MAC address?

A unique Layer 2 hardware address assigned to a network interface.

5. What is an IP address?

A logical Layer 3 address used for routing packets between networks.

6. Why do we need both MAC and IP addresses?

MAC addresses are used for communication within the local network, while IP addresses are used to identify and route devices across networks.

7. What is a port number?

A logical identifier used by the operating system to deliver network traffic to the correct application.

8. What is a protocol?

A defined set of communication rules that devices follow to exchange data correctly.

Day 1 Summary

After completing Day 1, you should confidently understand:

- ✔ What networking is and why it is important
- ✔ LAN, MAN, and WAN
- ✔ Client-Server basics
- ✔ NIC and its role
- ✔ MAC vs. IP addresses
- ✔ Hostnames and port numbers
- ✔ What protocols are
- ✔ Data units (Frame, Packet, Segment, Datagram)

- ✔ Essential Linux networking commands (`hostname` , `ip` , `ping`)
- ✔ Answers to common beginner networking interview questions

Next (Day 2): We'll dive into the **OSI Model**, explaining each of the seven layers in detail, how data moves through them, the protocols used at each layer, and how interviewers commonly ask OSI-related questions.